Project 2

Every student should submit the code, report (in pdf format) and ppt on Blackboard system before 23:59 June 2. Report can be written either in English or Chinese. Name your report as `studentID_Name.pdf` and your code as `studentID_name.zip`.

Q1. Data Agent System

Design and build a **generative** Data Agent system. A user without expert knowledge in programming or data science can directly obtain end-to-end data analysis pipelines, mathematical problem abstraction, and predictive models by prompting the LLM with natural questions. The training dataset is from DataMind-12K, a high-quality trajectory set specifically designed for Supervised Fine-Tuning (SFT) of data-analytic agents.

(1) [3 points] Read A Survey on Data Selection for Language Models (Data Selection for Instruction-Tuning and Multitask Training Part). Illustrate three methods in detail that you consider most effective for data selection within the data science and mathematical modeling domain.

(2) [3 points] Write python code to process data. Download the `datamind_12k.json` file from the DataMind-12K repository. Select 2k samples from this JSON file as training data, and 500 samples as validation data. Use the agent trajectory selection/filtering methods you illustrated in task (1) (such as complexity-based filtering, trajectory deduplication, or reward-based selection, etc.) rather than just randomly sampling. Prepare the selected data according to the requirements of Qwen model training. You may want to use free APIs from GLM-4.7-Flash.

(3) [3 points] Use Ray-train python code to train the Data Agent model by further finetuning the `Qwen3.5-0.8B` model. As no GPU is available in the server, you can use pytorch-cpu to debug your code, train the model for a few hours, and save a checkpoint. Note that the validation set is used for designing the hyperparameters and selecting the model checkpoint. You can also rent the GPU server in the AutoDL platform or use GPUs from google colab.

(4) [1 points] Deploy the agent model with your laptop and prepare a website for the demo. You can use the official `Qwen3.5-0.8B` checkpoint in this demo. You can refer to the tutorials here. Note that you may need to change the web UI to better support data analysis task.

Q2. Startup Business Plan

The advancement of large language models makes it possible that small teams can achieve significant business success in various startups, such as Midjourney and Meshy.

(1) [3 points] Suppose you plan to build a profitable startup focusing on LLM applications in the indie way (see reference here). Brainstorm the idea for a startup and write a business plan (see references here and here) with the help of LLM. Survey and compare your startup with the main competitors in the market. Also write a roadshow presentation ppt to secure funding.

You can refer to the ideas on Product Hunt as well as the suggestions on how to startup a company.

Please claim the LLM system you use in your report. You are responsible to check the content and make sure that the content is correct. Write all the questions/prompts you use to chat with the bot.

(2) [3 points] Prompt LLMs to design the system architecture of your product, to support industrial-grade deployment and 100,000-level concurrency. The system has modules like LLM engine, data processing, database, modules to support high concurrency, monitoring and operation Module, etc. Write an architecture design document with the system design diagram.

You can search references like AI-System-Design and System-Design-Primer and System-Design-101.

Presentation

[4 points] Presentations are on June 3 & June 4. Each sampled student has 10 minutes. 1~2 related questions will be asked after the presentation. Those who have not presented for Project 1 will have higher priority this time. The rest students will submit the recorded 10-minutes video for this presentation.